

Chapter 14

Magnetostatics

14.1 The vector potential

In magnetostatics, the Maxwell equations reduce to

$$\vec{\nabla} \cdot \vec{B} = 0 \quad , \quad \vec{\nabla} \times \vec{B} = \frac{4\pi}{c} \vec{J} \quad (14.1)$$

The first implies that we can write

$$\vec{B} = \vec{\nabla} \times \vec{A} \quad (14.2)$$

while taking the divergence of the second implies

$$\vec{\nabla} \cdot \vec{J} = 0 \quad (14.3)$$

Substituting (14.2) into Ampère's law gives

$$\vec{\nabla} \times (\vec{\nabla} \times \vec{A}) = \frac{4\pi}{c} \vec{J} \quad (14.4)$$

that is

$$\vec{\nabla} (\vec{\nabla} \cdot \vec{A}) - \vec{\nabla}^2 \vec{A} = \frac{4\pi}{c} \vec{J} \quad (14.5)$$

We drop the first term by choosing the Coulomb gauge $\vec{\nabla} \cdot \vec{A} = 0$ (we can always find a gauge parameter Λ to satisfy this condition), leading to

$$\vec{\nabla}^2 \vec{A} = -\frac{4\pi}{c} \vec{J} \quad (14.6)$$

This is the same as the Poisson equation for ϕ , but three copies of it – hence the same Green’s function. With $\vec{A} \rightarrow 0$ at infinity,

$$\vec{A} = \frac{1}{c} \int \frac{\vec{J}(\vec{x}')}{|\vec{x} - \vec{x}'|} dV' \quad (14.7)$$

14.2 Example: a current loop

Consider a circular loop of radius a carrying current I , as depicted in the Figure. The current density is

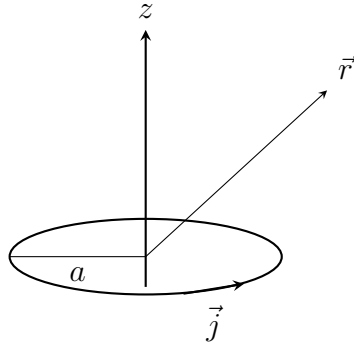


Figure 14.1: A circular loop of radius a carrying current I ; the current density $\vec{j}$ points along $\hat{\varphi}$.

$$\vec{J}(\vec{x}) = I \delta(\cos \theta) \frac{1}{a} \delta(r - a) \hat{\varphi} \quad , \quad \delta(\cos \theta) = \frac{1}{\sin \theta} \delta\left(\theta - \frac{\pi}{2}\right) \quad (14.8)$$

Note that the total current integrates to zero

$$\int \vec{J} dV = 2\pi a I \int \hat{\varphi} d\varphi = 0 \quad (14.9)$$

From (14.7),

$$\vec{A} = \frac{1}{c} \int \frac{j^\varphi(\vec{x}')}{|\vec{x} - \vec{x}'|} dV' \hat{\varphi} \quad (14.10)$$

but the unit vector $\hat{\varphi}$ changes around the loop, so we write the Cartesian components

$$A^x = -A^\varphi \sin \varphi \quad , \quad A^y = A^\varphi \cos \varphi \quad (14.11)$$

and

$$j^x = -j^\varphi \sin \varphi' \quad , \quad j^y = j^\varphi \cos \varphi' \quad (14.12)$$

Cylindrical symmetry allows us to take $\varphi = 0$, so that

$$A_y = A_\varphi = \frac{1}{c} \int \frac{j^\varphi \cos \varphi'}{|\vec{x} - \vec{x}'|} dV' \quad (14.13)$$

Using

$$\hat{r} = (\sin \theta \cos \varphi, \sin \theta \sin \varphi, \cos \theta) \quad , \quad \hat{r}' = (\sin \theta' \cos \varphi', \sin \theta' \sin \varphi', \cos \theta') \quad (14.14)$$

we have

$$|\vec{x} - \vec{x}'| = \sqrt{r^2 + r'^2 - 2 r r' (\sin \theta \sin \theta' \cos \varphi' + \cos \theta \cos \theta')} \quad (14.15)$$

so that

$$A_\varphi = \frac{I a}{c} \int \frac{\cos \varphi' d\varphi'}{\sqrt{r^2 + a^2 - 2 a r \sin \theta \cos \varphi'}} \quad (14.16)$$

Far from the loop,

$$r \gg a \Rightarrow A_\varphi \simeq \frac{I \pi a^2 \sin \theta}{c r^2} \quad (14.17)$$

The magnetic field follows from $\vec{B} = \vec{\nabla} \times \vec{A}$ in spherical coordinates

$$B_r = \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta A_\varphi) \quad , \quad B_\theta = -\frac{1}{r} \frac{\partial}{\partial r} (r A_\varphi) \quad , \quad B_\varphi = 0 \quad (14.18)$$

giving

$$r \gg a \Rightarrow B_r = 2 \frac{I \pi a^2 \cos \theta}{c r^3} \quad , \quad B_\theta = \frac{I \pi a^2 \sin \theta}{c r^3} \quad , \quad B_\varphi = 0 \quad (14.19)$$

14.3 The Biot-Savart law

More generally,

$$\begin{aligned} \vec{B} = \vec{\nabla} \times \vec{A} &= \vec{\nabla} \times \int \frac{\vec{J}(\vec{x}') dV'}{c |\vec{x} - \vec{x}'|} = \frac{1}{c} \int \vec{\nabla} \times \left(\frac{\vec{J}(\vec{x}')}{|\vec{x} - \vec{x}'|} \right) dV' \\ &= \frac{1}{c} \int -\vec{J}(\vec{x}') \times \vec{\nabla} \frac{1}{|\vec{x} - \vec{x}'|} dV' \\ &= \frac{1}{c} \int \vec{J}(\vec{x}') \times \frac{\vec{x} - \vec{x}'}{|\vec{x} - \vec{x}'|^3} dV' \end{aligned} \quad (14.20)$$

known as the Biot–Savart law. For a surface current we substitute

$$\vec{J}(\vec{x}') dV' \rightarrow \vec{K}(\vec{x}') \delta(z - z_0) dV' \equiv \vec{K}(\vec{x}') dA' \quad (14.21)$$

where $\vec{J}$ is current per unit area while $\vec{K}$ is current per unit length; for a line current

$$\vec{J}(\vec{x}') dV' \rightarrow I d\vec{s} \delta(r - r_0) dr \equiv I d\vec{s} \quad (14.22)$$

and for a point charge

$$\vec{J}(\vec{x}') dV' \rightarrow q \vec{v} \delta^3(\vec{r} - \vec{r}_0) dV' \quad (14.23)$$

Note also Ampère's law

$$\oint \vec{B} \cdot d\vec{s} = \frac{4\pi}{c} I[?] \quad (14.24)$$